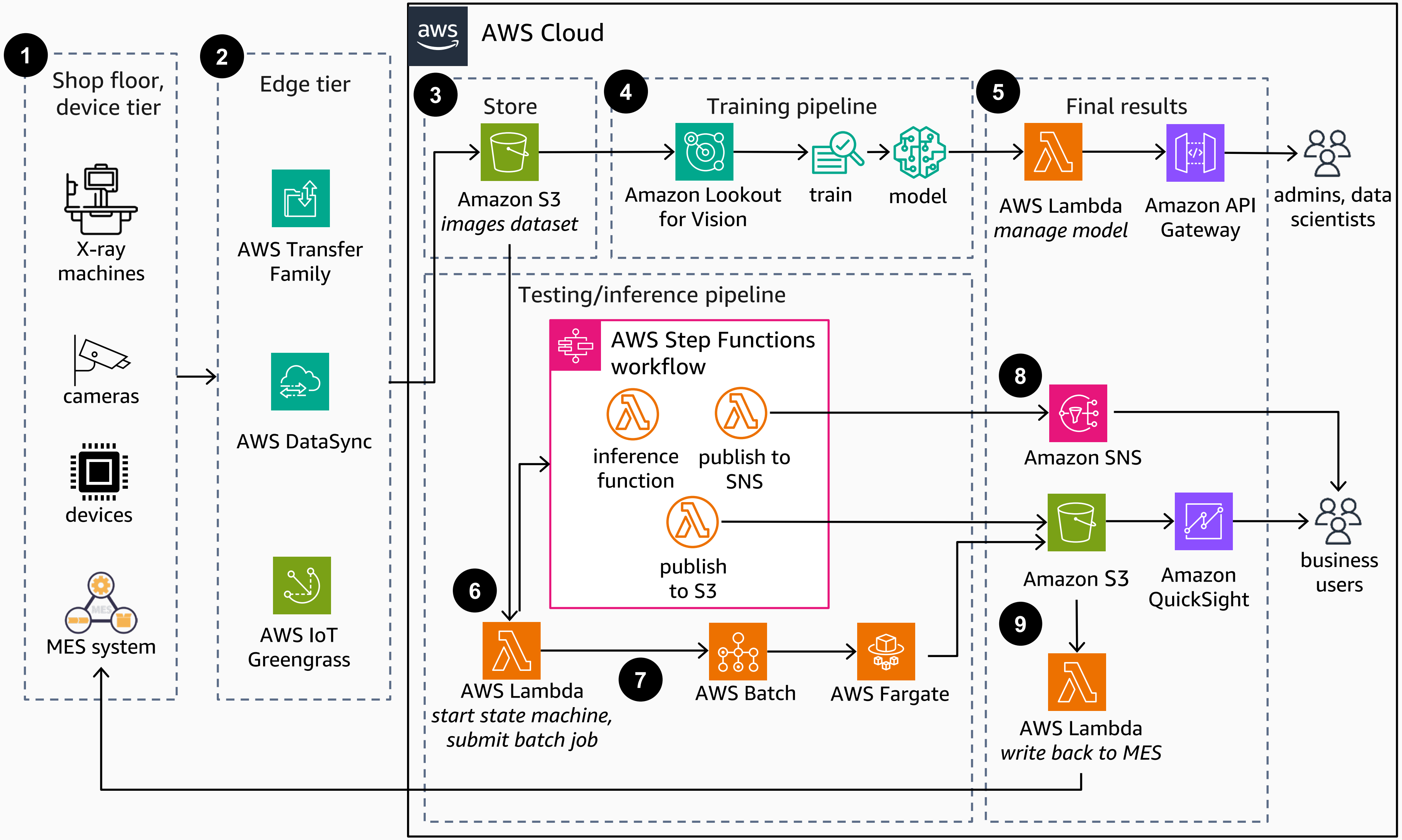
Identify Product Defects using Industrial Computer Vision

This reference architecture demonstrates how to detect anomalies such as casting metal defects, damage, and irregularities in X-ray images using Amazon Lookout for Vision, Amazon S3, and AWS Lambda for quality inspection in manufacturing.



1. Capture mages under consistent image capture conditions using X-ray machines, cameras, and other devices on the shop floor.
2. Transfer unique product images to AWS using **AWS Transfer Family** (transfer from systems relying on file transfer protocols), **AWS DataSync**, or **AWS IoT Greengrass** (to transfer images captured by edge devices).
3. Store product images on **Amazon Simple Storage Service** (Amazon S3) and separate into train and test datasets.
4. Use **Amazon Lookout for Vision** on the training dataset to automatically label, train, tune, and deploy the defect detection model.
5. The model is exposed through the **Amazon API Gateway** and **AWS Lambda**, which admins and data scientists can use to manage it.
6. For inference during runtime, use **AWS Lambda** to start the model for inference to detect anomalies. For a modular and scalable architecture, use **AWS Step Functions** with **Lambda** functions and to orchestrate a serverless workflow.
7. To run anomaly detection jobs in batch mode, submit a batch job to **AWS Batch**. Provision the necessary compute resources using **AWS Fargate**.
8. Notify the user on the confidence level using **Amazon Simple Notification Service** (Amazon SNS). If the minimum confidence is not met, the user must provide input to rightly label the undetected dataset.
9. The final results can be stored on **Amazon S3** to allow business users to visualize the results using **Amazon QuickSight**. The final results can also be written back into the manufacturing execution system (MES) on the shop floor.

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AWS Reference Architecture